**Department of Computer Science and Engineering**
**Academic Year 2021 - 2022 (Even Semester)**

**Degree, Semester & Branch:** VIII Semester B.E. CSE
**Course Code & Title:** GE8076 & Professional Ethics in Engineering
**Name of the Faculty member (s):** Dr.V.Anusuya, ASCP/CSE

## Innovative Practice Description

- **Unit / Topic:** Unit 3 / A Balanced Outlook on Law
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO11
- **Activity Chosen:** Think Pair Share
- **Justification:**
  - Think-pair-share is a cooperative learning technique.
  - The students should know the rules and regulations that control engineers, code of ethics, civil and criminal law, relationship between law and ethics.
  - This activity helps the students to understand the Law and Ethics and its benefits clearly.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
  - T : (Think) Teachers begin by asking a specific question about the text. Students "think" about what they know or have learned about the topic.
  - P : (Pair) Each student should be paired with another student or a small group.
  - S : (Share) Students share their thinking with their partner. Teachers expand the "share" into a whole-class discussion.
  - Step 1 (THINK) – in this step, students were asked to define a concept or term in the topic Balanced outlook on law. Then, they were asked to think about its meaning and write down its use in the society in a paper.
  - In Step 2 (PAIR) – students were asked to discuss with the individual sitting next to him or her about the terms and concepts identified in step1. The rationale in step 2 is to not only understand the concepts from each student's point of view but also learn from each other.
  - Step 3 (SHARE) involved sharing of the experience of learning the concept with the rest of the class. The rationale in step 3 is to understand the term or concept from a variety of perspectives.

- The students pairs are Kavimalar and Samyukthavikashini,Mohan kumar and Abishek,Gnamoorthy vidhya and Nivethitha,Vijayalakshmi and Karthiga,Kannan and Mohamed Shein. They presented the topic Code of ethics, Balanced outlook on law, law and ethics as shown in figure1 and figure2.

- **CO – PO / PSO mapping:**

| CO | PO1 | PO2 | PO6 | PO8 | PO9 | PO12 | PSO1 | PSO2 | PSO3 |
|---|---|---|---|---|---|---|---|---|---|
| **CO1** | 1 | 1 | 3 | 3 | 3 | 1 | 1 | 1 | 1 |

**(1 – Low 2 – Moderate 3 – High)**

- **PO / PSO mapped:**

| **Innovative practice** | **PO1** | **PO2** | **PO6** | **PO8** | **PO9** | **PO12** | **PSO1** | **PSO2** | **PSO3** |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1 | 3 | 3 | 3 | 1 | 1 | 1 | 1 |
| **Justification for correlation** | Needs to understand the role of engineers as experimenters, fundamental Engineering knowledge is essential | The students will study about engineering experiments, they could gain problem analysis skills | The students can identify the role of engineers and code of ethics required to protect public and public interest at the local, regional and global level. | The students can identify situations of unethical professional conduct and propose ethical alternatives as they know roles of engineers and code of ethics one has to follow. | As students are learning roles of engineers as managers, experimenters they can handle a team and team members | Consciousness about the role of engineers as managers, witness and advisors for continuous development of profession should follow the code of ethics. | As knowledge about the code of ethics are required to be a better software developer. | To apply the knowledge of IoT, Cloud computing and cybersecurity, they should know the code of ethics during development | To provide a solution to real world problem using Artificial Intelligence and bigdata they should learn code of ethics. |

- **Images / Screenshot of the practice:**

***Figure 1: Think Pair Share Implementation***

***Figure 2: Think Pair Share Implementation***

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - All the students were actively participated and they understood the concepts correctly.
    - Students explored the knowledge of the rules and regulations that control engineers, code of ethics.
    - It makes the students to understand the civil and criminal law, relationship between law and ethics.
  - ***Benefit of the practice:***
    - Students can able to understand the details about balanced outlook on law

- Students can able to gain knowledge about the code of ethics are required to be a better software developer This activity helps the students to know the importance of ethics during software development.

- ***Challenges faced in implementation:***
    - Two team members Mohankumar and Kannan presented the same topic. The Samyuktha vikashini and Vijayalakshmi teams are presented code of ethics in traffic signal and software industry case study.

**References:**


- https://www.ritrjpm.ac.in/images/computer-science/
- https://www.readingrockets.org/sites/default/files/Think-Pair-Share-template.pdf
- https://www.readingrockets.org/strategies/think-pair-share


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| **Signature of Faculty Member** | **HOD** |
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